LECTURE 3

3 Fourier series

TUESDAY 6 OCTOBER

Last week we separated variables in the heat equation on a ring and found solutions of the form $e^{-n^2 t}\cos nx$ and $e^{-n^2 t}\sin nx$. Any finite sum of these is again a solution, so the obvious question is whether *every* initial temperature profile can be written as such a sum. Fourier claimed in 1807 that it can, provided the sum is allowed to be infinite.[1] Today we make that claim precise for 2π-periodic functions and compute our first series.

[1] The claim met resistance from Lagrange, who doubted that a discontinuous function could be a sum of smooth ones. Both were partly right; see §3.3.

3.1 Periodic functions and an inner product

A function $f:\mathbb{R}\to\mathbb{R}$ is *2π-periodic* if $f(x+2\pi)=f(x)$ for all x. Such a function is determined by its values on any interval of length 2π; we use $[-\pi,\pi)$ throughout. Geometry enters through the following pairing, which behaves like a dot product with infinitely many coordinates.

DEFINITION 3.1 Inner product For integrable 2π-periodic functions f and g, set

$$\langle f,g\rangle=\frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\,g(x)\,dx, \qquad \|f\|=\langle f,f\rangle^{1/2}.$$

We call f and g *orthogonal* when $\langle f,g\rangle=0$.

Some texts use $\frac{1}{2\pi}$ instead; the coefficient formulas then change by a factor of two. Check which convention a source uses before copying a result.

The factor $1/\pi$ is a convention chosen so that $\cos nx$ and $\sin nx$ have norm one. The next lemma is the whole reason the trigonometric functions are a good coordinate system.

LEMMA 3.2 Orthogonality *For integers $m,n\geq 1$,*

$$\langle \cos mx,\cos nx\rangle=\langle \sin mx,\sin nx\rangle=\delta_{mn}, \qquad \langle \cos mx,\sin nx\rangle=0,$$

and $\langle 1,1\rangle=2$, while 1 is orthogonal to every $\cos nx$ and $\sin nx$.

Proof. Use the product formula $2\cos mx\cos nx=\cos(m-n)x+\cos(m+n)x$. The integral of $\cos kx$ over a full period vanishes unless $k=0$, which

happens only for the first term when $m = n$. The other two cases follow in the same way from the corresponding product formulas. ∎

3.2 Fourier coefficients

Suppose for a moment that f really is a sum $\frac{a_0}{2} + \sum_{n\ge 1}(a_n \cos nx + b_n \sin nx)$ and that we may integrate term by term. Taking the inner product with $\cos mx$ kills every term except one, by Lemma 3.2. This forces the coefficients and motivates the definition.

DEFINITION 3.3 Fourier series The *Fourier coefficients* of an integrable 2π-periodic f are

$$a_n = \langle f, \cos nx\rangle, \qquad b_n = \langle f, \sin nx\rangle, \qquad n \ge 0,$$

and its *Fourier series* is the formal sum

$$f(x) \sim \frac{a_0}{2} + \sum_{n=1}^{\infty}\big(a_n \cos nx + b_n \sin nx\big). \tag{1}$$

The Nth *partial sum* $S_N f$ stops the sum at $n = N$.

The symbol $\sim$ is deliberate: nothing yet says the series converges, let alone to f.[2] Two shortcuts save most of the work in practice. If f is even, every b_n vanishes; if f is odd, every a_n vanishes.

[2] Lecture 4 answers both questions for piecewise smooth functions.

EXAMPLE 3.4 Square wave Let $f(x) = \operatorname{sgn}(x)$ on $(-\pi, \pi)$, extended periodically. Compute its Fourier series.

Solution. The function is odd, so $a_n = 0$ for all n. For the sine coefficients,

$$b_n = \frac{2}{\pi}\int_0^{\pi} \sin nx\,dx = \frac{2}{\pi}\cdot\frac{1-\cos n\pi}{n} = \begin{cases} \dfrac{4}{\pi n}, & n \text{ odd},\\[2mm] 0, & n \text{ even}. \end{cases}$$

Hence $\operatorname{sgn}(x) \sim \frac{4}{\pi}\big(\sin x + \frac{1}{3}\sin 3x + \frac{1}{5}\sin 5x + \cdots\big)$. ∎

The coefficients decay only like $1/n$, which is slow. That is typical of a function with a jump, and Figure 3.1 shows what it costs.

3.3 The Gibbs phenomenon

Look closely at the peaks next to $x = 0$ in Figure 3.1. As N grows the overshoot moves closer to the jump but does not shrink. One can show

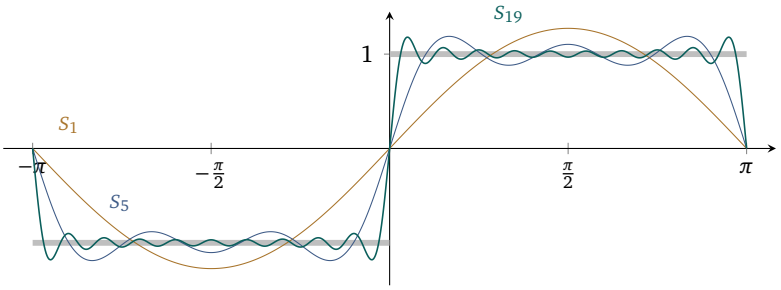


FIGURE 3.1 Partial sums of the square-wave series. Away from the jump they settle down; next to it each S_N overshoots by about 9% of the jump.

that

$$\lim_{N\to\infty} \max_x S_N f(x) = \frac{2}{\pi}\int_0^{\pi} \frac{\sin t}{t}\,dt \approx 1.1790,$$

an overshoot of roughly 9% of the total jump of 2.[3]

[3] Named after J. W. Gibbs, who described it in 1899, although Wilbraham had found it fifty years earlier.

KEY IDEA

Smoothness of f and decay of its Fourier coefficients are two views of the same thing. A jump gives coefficients of size $1/n$ and a persistent overshoot; each extra continuous derivative buys one more power of $1/n$.

Computing partial sums. The overshoot is easy to see numerically. The script below evaluates $S_N f$ on a fine grid and reports the maximum; try doubling `N` a few times.

```python
import numpy as np

def square_wave_partial_sum(x, N):
    """S_N of sgn(x): only odd harmonics contribute."""
    n = np.arange(1, N + 1, 2)              # 1, 3, 5, ...
    return 4 / np.pi * np.sin(np.outer(x, n)) @ (1 / n)

x = np.linspace(1e-4, 0.5, 20_000)
for N in (19, 99, 499):
    print(N, square_wave_partial_sum(x, N).max())   # -> about 1.179
```

Table 3.1 collects a few series that recur in the problem sets. Each follows from Definition 3.3 with one or two integrations by parts.

$f(x)$ on $(-\pi,\pi)$	Fourier series	Decay
$\operatorname{sgn} x$	$\frac{4}{\pi}\sum_{n\ \text{odd}} \frac{\sin nx}{n}$	$1/n$
x	$2\sum_{n\ge 1}(-1)^{n+1}\frac{\sin nx}{n}$	$1/n$
$\lvert x\rvert$	$\frac{\pi}{2} - \frac{4}{\pi}\sum_{n\ \text{odd}} \frac{\cos nx}{n^2}$	$1/n^2$
x^2	$\frac{\pi^2}{3} + 4\sum_{n\ge 1}(-1)^n\frac{\cos nx}{n^2}$	$1/n^2$

TABLE 3.1 Common series. Continuous periodic extensions ($|x|$, x^2) decay faster than those with a jump.

3.4 The complex form

Euler's formula $e^{inx} = \cos nx + i \sin nx$ packs each pair of sine and cosine terms into two exponentials. The bookkeeping becomes simpler, and the formulas extend without change to complex-valued f.[4]

[4] This is the form used in signal processing and in the discrete Fourier transform, where the integral becomes a finite sum over samples.

DEFINITION 3.5 Complex coefficients For an integrable 2π-periodic f and every integer n, set

$$c_n = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x)\, e^{-inx}\, dx, \qquad f(x) \sim \sum_{n=-\infty}^{\infty} c_n\, e^{inx}.$$

The two descriptions carry the same information: $c_0 = a_0/2$ and $c_{\pm n} = (a_n \mp i b_n)/2$ for $n \geq 1$. For real f the coefficients satisfy $c_{-n} = \overline{c_n}$, so half of them suffice.

3.5 Exercises

EXERCISE 3.6 Show that if f is even then $b_n = 0$ for every n, and that $a_n = \frac{2}{\pi} \int_0^{\pi} f(x) \cos nx\, dx$.

EXERCISE 3.7 Triangle wave Derive the series for $|x|$ in Table 3.1. Then differentiate it term by term and compare the result with Example 3.4. Why is the differentiation legitimate here but not for the square wave?

EXERCISE 3.8 Shifts Let $g(x) = f(x - h)$. Express the complex coefficients of g in terms of those of f, and explain why $|c_n|$ does not depend on h.

LECTURE 4

4 Convergence and Parseval

THURSDAY 8 OCTOBER

We now know how to compute a Fourier series. This lecture answers the two questions left open in Lecture 3: when does $S_N f(x)$ converge, and to what? The main tool is a closed form for the partial sums.

4.1 The Dirichlet kernel

Substituting the coefficient formulas into $S_N f$ and collecting the cosines gives an integral against a single function,

$$S_N f(x) = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x-t)\, D_N(t)\, dt, \tag{2}$$

where

$$D_N(t) = \sum_{n=-N}^{N} e^{int} = \frac{\sin\big((N+\tfrac{1}{2})t\big)}{\sin(t/2)}.$$

The closed form is a geometric sum.[5] The kernel has total integral 2π and a tall central peak of height $2N+1$, so $S_N f(x)$ is a weighted average of f near x.

[5] Multiply the sum by $e^{it/2} - e^{-it/2}$; everything telescopes except two terms.

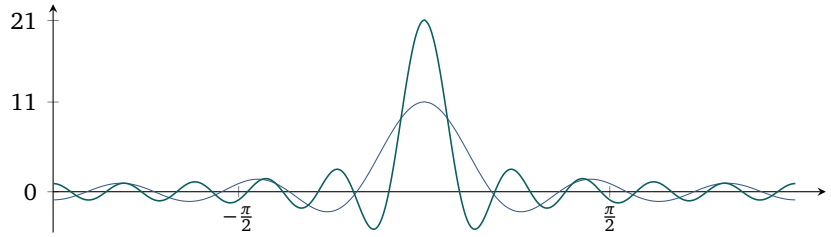


FIGURE 4.2 The Dirichlet kernels D_5 and D_{10}. The peak grows but the side lobes do not die out, which is why D_N is not a good averaging kernel.

Unfortunately D_N takes negative values and its absolute integral grows like $\log N$, so (2) is not an honest average. Convergence needs some regularity of f near x.

4.2 Pointwise convergence

THEOREM 4.9 Dirichlet *Let f be 2π-periodic and piecewise C^1. Then for every x,*

$$\lim_{N\to\infty} S_N f(x) = \frac{f(x^+) + f(x^-)}{2}.$$

In particular $S_N f(x) \to f(x)$ wherever f is continuous.

Proof. Write $S_N f(x) - f(x^+)$ as an integral of D_N against $g(t) = \big(f(x-t) - f(x^+)\big)/\sin(t/2)$ over $(-\pi, 0)$, and similarly for $f(x^-)$. Piecewise smoothness makes g bounded near $t = 0$, and the Riemann–Lebesgue lemma sends both integrals to zero. ∎

For the square wave this predicts $S_N f(0) \to 0$, the midpoint of the jump, which agrees with every partial sum in Figure 3.1.

Note the difference from the Gibbs overshoot: the *value* at each fixed x converges, but the *maximum* near the jump does not. Convergence is pointwise, not uniform.

4.3 Parseval's identity

The orthogonality relations make the coefficients behave like coordinates in an orthonormal basis. The analogue of Pythagoras' theorem holds with infinitely many terms.

THEOREM 4.10 Parseval *If f is square integrable on $[-\pi, \pi]$, then*

$$\frac{1}{\pi}\int_{-\pi}^{\pi} |f(x)|^2\,dx = \frac{a_0^2}{2} + \sum_{n=1}^{\infty}\big(a_n^2 + b_n^2\big).$$

KEY IDEA

Energy is conserved between the two descriptions of f: the mean square of the function equals the sum of the squares of its coefficients.

EXAMPLE 4.11 The Basel problem Use Parseval's identity for $f(x) = x$ to evaluate $\sum_{n\ge 1} 1/n^2$.

Solution. From Table 3.1, $a_n = 0$ and $b_n = 2(-1)^{n+1}/n$. The left side is $\frac{1}{\pi}\int_{-\pi}^{\pi} x^2\,dx = \frac{2\pi^2}{3}$. Parseval gives

$$\frac{2\pi^2}{3} = \sum_{n=1}^{\infty}\frac{4}{n^2} \qquad\Longrightarrow\qquad \sum_{n=1}^{\infty}\frac{1}{n^2} = \frac{\pi^2}{6}.$$

This is Euler's 1734 result, here obtained in three lines. ∎

4.4 Back to the heat equation

We can now finish the problem that opened Lecture 3. The temperature $u(x,t)$ on a ring of circumference 2π satisfies $u_t = u_{xx}$ with $u(x,0) = f(x)$. Expanding f as in (1) and evolving each mode separately,

$$u(x,t) = \frac{a_0}{2} + \sum_{n=1}^{\infty} e^{-n^2 t}\big(a_n\cos nx + b_n \sin nx\big).$$

The factor $e^{-n^2 t}$ damps high frequencies fastest, so for every $t > 0$ the coefficients decay faster than any power of n and u is smooth, even when f is the square wave.[6]

[6] This instant smoothing is special to diffusion. The wave equation, in Lecture 6, keeps jumps forever.

EXAMPLE 4.12 Cooling a half-heated ring The ring starts with temperature 1 on $(0, \pi)$ and -1 on $(-\pi, 0)$. How quickly does it approach equilibrium?

Solution. By Example 3.4 only odd sine modes are present, so

$$u(x,t) = \frac{4}{\pi} \sum_{n \text{ odd}} \frac{e^{-n^2 t}}{n} \sin nx.$$

For $t \geq 1$ the first term dominates, since the next one is smaller by a factor $\frac{1}{3} e^{-8t} < 10^{-4}$. The ring therefore relaxes to the mean temperature 0 like $\frac{4}{\pi} e^{-t} \sin x$: one smooth bump that fades at unit rate. ∎

4.5 Exercises

EXERCISE 4.13 Apply Parseval's identity to $f(x) = x^2$ and deduce that $\sum_{n \geq 1} 1/n^4 = \pi^4/90$.

EXERCISE 4.14 Uniform convergence Show that if f is continuous, 2π-periodic and piecewise C^1, then $\sum_n (|a_n| + |b_n|) < \infty$. Conclude that $S_N f \to f$ uniformly. Which hypothesis fails for the square wave?